MEGATHON 2026 · DELIVERABLE M2 (MID-SUBMISSION 2· 4:00 AM, 11TH OCTOBER)

SAHAAYA: "Every Need. Every Voice. Every Me"

1. Current Approach and Changes Since M1

SAHAAYA has transitioned from conceptual design into a functional, end-to-end Progressive Web App (PWA). The system guides users across every active phase of restaurant dining—from menu parsing and item ordering to real-time staff communication, bill auditing, and feedback logging. To ensure reliable performance on standard consumer hardware, practical implementation during the midnight sprint drove three critical architectural evolutions:

Sequential Audio-Visual Flow Control: Initial live builds encountered microphone acoustic feedback loops where the assistant picked up its own synthesized speech. We engineered a strict sequential execution state machine: all page and item read-aloud TTS sequences now run to total completion before unlocking and activating the Automatic Speech Recognition (ASR) microphone dock.

OCR Normalization & Schema Mapping: Unstructured physical menu layouts (multi-column lists, stylized cursive scripts, and separated price grids) initially created parsing discrepancies. We upgraded the OCR normalization layer to automatically clean bounding-box text and map extracted items directly to structured dietary attributes (Veg, Non-Veg, Jain, Allergens) and verified pricing models before client rendering.

Resilience & Local Developer Simulation Tooling: To accelerate testing without constant network restarts, we embedded client-side simulation controls. Engineers can directly inject synthetic network drops and simulate low-confidence ASR inputs, validating our planned recovery flows in real time.

2. Experiments Run So Far and Their Results

Between M1 and M2, three targeted technical experiments were conducted to evaluate parsing robustness, client-to-staff relay speed, and acoustic failure handling:

Experiment Track	Methodology & Scope	Observed Baseline & Challenge	Empirical Outcome / Solution
Exp 1: Multilingual OCR Extraction	Evaluated physical menu cards captured across English, Hindi, and Telugu under varying angles.	Naive heuristic extraction misaligned 38% of section boundaries and prices on artistic menus.	Structured parsing models normalized extracted text, flagged low-confidence prices, and triggered user confirmation.
Exp 2: Real-Time WebSocket Staff Relay	Benchmarked bidirectional communication over the /relay endpoint between diner PWA and staff dashboard.	Monitored payload latency under simulated constrained 3G / throttled Wi-Fi network conditions.	Average transmission latency remained under real-time thresholds, ensuring instant live-caption rendering on staff tablets.
Exp 3: ASR Noise Tolerance & Fallbacks	Tested voice command recognition against elevated ambient restaurant background babble.	Standard VAD caused speech truncation or misrecognitions when confidence scores dipped below 0.65.	Low-confidence scores successfully triggered fallback visual indicator panels and keypad selection modals.

3. Progress on Testing

Testing has moved from standalone unit checks to complete functional validation across our accessibility workflows, language mappings, and system error loops:

Condition-Specific Workflows: Validated end-to-end user journeys for all target profiles. The Visual Impairment track successfully navigates via camera scan-to-read and audio prompts. The Hearing Impairment track operates with zero audio dependencies via live transcripts, status cues, and waiter cards. The Speech/Motor track enables full meal selection through the dedicated phrase board and assistive TTS outputs.

Multilingual Language Integration: Tested speech and text input/output pipelines across English, Hindi, and Telugu. Verified cross-language phonetic mapping for native culinary terminology, confirming rapid transcription-to-speech round trips without semantic corruption.

OCR Parsing Under Suboptimal Lighting: Subjected the scanning pipeline to clear, dim, and tilted lighting environments. The parser reliably detects low-contrast fields and prompts the user for verbal or tap verification rather than inserting speculative values.

Staff Relay & Error Recovery Safeguards: Conducted rigorous failure simulation tests covering low-confidence ASR events and sudden network disconnections. Critical actions—specifically cart checkouts, order submissions, and payment requests—are strictly gated behind explicit confirmation modals, eliminating accidental orders.

4. Plan for the Remaining Time (4:00 AM – 10:00 AM)

The remaining development and packaging sprint focuses on user experience polish, code governance, and pitch execution:

Action Area	Target Scope & Deliverables	Execution Window
Accessibility Polish	Finalize WCAG AAA high-contrast visual themes, responsive text scaling, and adjustable voice playback speed controls for elderly and motor profiles.	04:00 AM – 06:00 AM
Repository & Docs Cleanup	Organize codebase modules, document environment configurations and API proxy routes, and maintain clear setup guidelines for team collaboration.	06:00 AM – 07:30 AM
Demo Readiness & Dry-Runs	Conduct mock dry-runs of the live guest-to-staff relay and Bhashini speech exchange; record the 4-minute captioned video and finalize presentation slides.	07:30 AM – 09:30 AM